

LEVEL 6 · WRITE-UP

The Last Door

Five physics problems each yield a small integer key. The five keys build five large primes; the primes are the factors of the sealed modulus in core.sealed; the modulus decrypts the flag. Nothing the player receives names the method — the three unlabeled numbers (a modulus, 65537, a ciphertext) must be recognised and assembled.

The five keys

Door I	Schwarzschild	: $r_s = 2GM/c^2$, $M=10 M_{\text{sun}}$	-> 29538 m	-> 29538
Door II	Jeans mass	: $20 K$, $n=1e3/cm^3$, $\mu=2.3$	-> 49 M_{sun}	-> 49
Door III	Hubble	: $v=c z$ ($z=0.030$), $d=v/H_0$ (70)	-> 128 Mpc	-> 128
Door IV	orbit (Kepler)	: $a=4.0 \text{ AU}$, $M=1 M_{\text{sun}}$	-> 2922 days	-> 2922
Door V	Fourier	: dominant tone of carrier.wav	-> 1420 Hz	-> 1420

Door-by-door

```
I   r_s = 2*6.674e-11*(10*1.989e30)/(2.998e8)^2 = 29538.45 -> 29538
II  rho=mu*m_p*n; M_J=(5kT/G mu m_p)^1.5 (3/4*pi*r)^0.5 /M_sun ≈ 49
III v = c*z = 2.998e5*0.030 = 8994 km/s; d = 8994/70 = 128.5 -> 128
IV  T = 2*pi/(a^3/GM)/86400, a=4 AU = 5.984e11 m ≈ 2922 days
V   FFT carrier.wav -> peak at 1420 Hz (21 cm H line)
```

Opening the core

```
from sympy import nextprime
keys = [29538, 49, 128, 2922, 1420]      # door order I..V
OFF = [0xA1, 0xB2, 0xC3, 0xD4, 0xE5]
primes=[nextprime(S*10**60 + o*10**50 + 7) for S,o in zip(keys,OFF)]
n,e,c =
assert primes[0]*primes[1]*primes[2]*primes[3]*primes[4] == n
phi=1
for p in primes: phi*=(p-1)
d=pow(e,-1,phi); m=pow(c,d,n)
print(m.to_bytes((m.bit_length()+7)//8,'big').decode())
```

FLAG{Kill_My_Pain}

Builder notes

- Ship core.sealed and carrier.wav. Keep solve.py and this write-up private.
- The story/prompt text is kept separate (prompt.txt) so you can edit it freely before pasting it into the CTFd challenge description above the doors.
- Pinned constants make the five keys exact: 29538, 49, 128, 2922, 1420.

- Flag exact-match, case-sensitive: FLAG{Kill_My_Pain}.
- Test the full solver on a fresh machine before the event.